

SQL Date Functions Exercise

BRIGHTLEARN

KHUTSO JOSHUA MONYEBODI

Introduction

This exercise is designed to build proficiency in SQL date functions across five progressive skill areas: extracting components from date values, converting between text and date data types, performing date arithmetic and interval calculations, filtering datasets by temporal criteria, and applying conditional logic to date-derived values. Through twenty structured questions plus a bonus challenge, learners will work with realistic business scenarios and order processing, customer retention, subscription management, and financial reporting to develop practical fluency rather than purely theoretical knowledge.

Question 1: Day Name from Order Date

```
6  -- Q1: Day Name from Order Date
7  CREATE OR REPLACE TABLE orders (
8      order_id    INT,
9      customer_id INT,
10     order_date   DATE
11 );
12 INSERT INTO orders VALUES
13     (1001, 101, '2026-05-01'),
14     (1002, 102, '2026-05-02'),
15     (1003, 103, '2026-05-03'),
16     (1004, 104, '2026-05-04'),
17     (1005, 105, '2026-05-05');
18
19 SELECT
20     order_id,
21     customer_id,
22     order_date,
23     DAYNAME(order_date) AS day_name
24 FROM orders;
```

Add parameter

← Results 3 of 3 →		Table	+		
	¹ ₃ order_id	¹ ₃ customer_id	 order_date	^A _C day_name	
1	1001	101	2026-05-01	Fri	
2	1002	102	2026-05-02	Sat	
3	1003	103	2026-05-03	Sun	
4	1004	104	2026-05-04	Mon	
5	1005	105	2026-05-05	Tue	

Question 2: Month Name from Signup Date

```
26 -- Q2: Month Name from Signup Date
27 CREATE OR REPLACE TABLE customer_signups (
28   customer_id INT,
29   customer_name STRING,
30   signup_date DATE
31 );
32 INSERT INTO customer_signups VALUES
33   (201, 'John', '2026-01-15'),
34   (202, 'Mary', '2026-02-20'),
35   (203, 'Peter', '2026-03-05'),
36   (204, 'Sarah', '2026-04-18'),
37   (205, 'Thabo', '2026-05-30');
38
39 SELECT
40   customer_id,
41   customer_name,
42   signup_date,
43   MONTHNAME(signup_date) AS signup_month_name
44 FROM customer_signups;
```

Add parameter

Results 3 of 3				
Table				
	customer_id	customer_name	signup_date	signup_month_name
1	201	John	2026-01-15	Jan
2	202	Mary	2026-02-20	Feb
3	203	Peter	2026-03-05	Mar
4	204	Sarah	2026-04-18	Apr
5	205	Thabo	2026-05-30	May

Question 3: Month Number from Sale Date

```
46 -- Q3: Month Number from Sale Date
47 CREATE OR REPLACE TABLE sales (
48   sale_id STRING,
49   product_name STRING,
50   sale_date DATE,
51   amount DECIMAL(10,2)
52 );
53 INSERT INTO sales VALUES
54   ('S001', 'Laptop', '2026-01-10', 12000),
55   ('S002', 'Mouse', '2026-02-15', 250),
56   ('S003', 'Keyboard', '2026-03-20', 700),
57   ('S004', 'Monitor', '2026-04-25', 3500),
58   ('S005', 'Desk', '2026-05-30', 2000);
59
60 SELECT
61   sale_id,
62   product_name,
63   sale_date,
64   MONTH(sale_date) AS sale_month
65 FROM sales;
```

Add parameter

Results 3 of 3				
Table				
	sale_id	product_name	sale_date	sale_month
1	S001	Laptop	2026-01-10	1
2	S002	Mouse	2026-02-15	2
3	S003	Keyboard	2026-03-20	3
4	S004	Monitor	2026-04-25	4
5	S005	Desk	2026-05-30	5

Question 4: Year from Transaction Date

```
67 -- Q4: Year from Transaction Date
68 CREATE OR REPLACE TABLE transactions (
69   transaction_id STRING,
70   customer_id INT,
71   transaction_date DATE,
72   amount DECIMAL(10,2)
73 );
74 INSERT INTO transactions VALUES
75   ('T001', 101, '2024-12-15', 500),
76   ('T002', 102, '2025-01-20', 1200),
77   ('T003', 103, '2025-06-10', 800),
78   ('T004', 104, '2026-02-05', 1500),
79   ('T005', 105, '2026-05-25', 2000);
80
81 SELECT
82   transaction_id,
83   customer_id,
84   transaction_date,
85   YEAR(transaction_date) AS transaction_year
86 FROM transactions;
```

Add parameter

← Results 3 of 3 →

Table +

	transaction_id	customer_id	transaction_date	transaction_year
1	T001	101	2024-12-15	2024
2	T002	102	2025-01-20	2025
3	T003	103	2025-06-10	2025
4	T004	104	2026-02-05	2026
5	T005	105	2026-05-25	2026

Question 5: Day of Month from Delivery Date

```
88 -- Q5: Day of Month from Delivery Date
89 CREATE OR REPLACE TABLE deliveries (
90   delivery_id STRING,
91   customer_id INT,
92   delivery_date DATE
93 );
94 INSERT INTO deliveries VALUES
95   ('D001', 101, '2026-05-01'),
96   ('D002', 102, '2026-05-08'),
97   ('D003', 103, '2026-05-15'),
98   ('D004', 104, '2026-05-22'),
99   ('D005', 105, '2026-05-29');
100
101 SELECT
102   delivery_id,
103   customer_id,
104   delivery_date,
105   DAY(delivery_date) AS day_of_month
106 FROM deliveries;
```

Add parameter

← Results 3 of 3 →

Table +

	delivery_id	customer_id	delivery_date	day_of_month
1	D001	101	2026-05-01	1
2	D002	102	2026-05-08	8
3	D003	103	2026-05-15	15
4	D004	104	2026-05-22	22
5	D005	105	2026-05-29	29

Question 6: Add Today's Date

```
111
112 -- Q6: Add Today's Date
113 CREATE OR REPLACE TABLE employees (
114     employee_id INT,
115     employee_name STRING,
116     department STRING
117 );
118 INSERT INTO employees VALUES
119     (301, 'Nandi', 'Sales'),
120     (302, 'Brian', 'IT'),
121     (303, 'Lerato', 'Finance'),
122     (304, 'Sipho', 'HR'),
123     (305, 'Aisha', 'Marketing');
124
125 SELECT
126     employee_id,
127     employee_name,
128     department,
129     CURRENT_DATE() AS today_date
130 FROM employees;
131
```

Add parameter

← Results 3 of 3 → Table +

	employee_id	employee_name	department	today_date
1	301	Nandi	Sales	2026-08-01
2	302	Brian	IT	2026-08-01
3	303	Lerato	Finance	2026-08-01
4	304	Sipho	HR	2026-08-01
5	305	Aisha	Marketing	2026-08-01

Question 7: Convert Text to Date

```
131
132 -- Q7: Convert Text to Date
133 CREATE OR REPLACE TABLE online_orders (
134     order_id INT,
135     customer_id INT,
136     order_date_text STRING
137 );
138 INSERT INTO online_orders VALUES
139     (4001, 101, '2026-01-15'),
140     (4002, 102, '2026-02-20'),
141     (4003, 103, '2026-03-25'),
142     (4004, 104, '2026-04-10'),
143     (4005, 105, '2026-05-05');
144
145 SELECT
146     order_id,
147     customer_id,
148     order_date_text,
149     TO_DATE(order_date_text) AS order_date
150 FROM online_orders;
151
```

Add parameter

← Results 3 of 3 → Table +

	order_id	customer_id	order_date_text	order_date
1	4001	101	2026-01-15	2026-01-15
2	4002	102	2026-02-20	2026-02-20
3	4003	103	2026-03-25	2026-03-25
4	4004	104	2026-04-10	2026-04-10
5	4005	105	2026-05-05	2026-05-05

Question 8: Format Date as Text

```
152 -- Q8: Format Date as Text
153 CREATE OR REPLACE TABLE payment_dates (
154   payment_id STRING,
155   customer_id INT,
156   payment_date DATE
157 );
158 INSERT INTO payment_dates VALUES
159   ('P001', 101, '2026-01-05'),
160   ('P002', 102, '2026-02-10'),
161   ('P003', 103, '2026-03-15'),
162   ('P004', 104, '2026-04-20'),
163   ('P005', 105, '2026-05-25');
164
165 SELECT
166   payment_id,
167   customer_id,
168   payment_date,
169   DATE_FORMAT(payment_date, 'yyyy-MM-dd') AS formatted_payment_date
170 FROM payment_dates;
```

Add parameter

Results 3 of 3				
	payment_id	customer_id	payment_date	formatted_payment_date
1	P001	101	2026-01-05	2026-01-05
2	P002	102	2026-02-10	2026-02-10
3	P003	103	2026-03-15	2026-03-15
4	P004	104	2026-04-20	2026-04-20
5	P005	105	2026-05-25	2026-05-25

Question 9: Days Since Last Purchase

```
171
172 -- Q9: Days Since Last Purchase
173 CREATE OR REPLACE TABLE customer_purchases (
174   customer_id INT,
175   customer_name STRING,
176   last_purchase_date DATE
177 );
178 INSERT INTO customer_purchases VALUES
179   (501, 'John', '2026-05-01'),
180   (502, 'Mary', '2026-05-10'),
181   (503, 'Peter', '2026-05-15'),
182   (504, 'Sarah', '2026-05-20'),
183   (505, 'Thabo', '2026-05-25');
184
185 SELECT
186   customer_id,
187   customer_name,
188   last_purchase_date,
189   DATEDIFF(CURRENT_DATE(), last_purchase_date) AS days_since_last_purchase
190 FROM customer_purchases;
```

Add parameter

Results 3 of 3				
	customer_id	customer_name	last_purchase_date	days_since_last_purchase
1	501	John	2026-05-01	92
2	502	Mary	2026-05-10	83
3	503	Peter	2026-05-15	78
4	504	Sarah	2026-05-20	73
5	505	Thabo	2026-05-25	68

Question 10: Expected Delivery Date

```
191
192 -- Q10: Expected Delivery Date
193 CREATE OR REPLACE TABLE shipping_orders (
194     order_id INT,
195     customer_id INT,
196     order_date DATE
197 );
198 INSERT INTO shipping_orders VALUES
199     (6001, 101, '2026-05-01'),
200     (6002, 102, '2026-05-03'),
201     (6003, 103, '2026-05-05'),
202     (6004, 104, '2026-05-07'),
203     (6005, 105, '2026-05-09');
204
205 SELECT
206     order_id,
207     customer_id,
208     order_date,
209     DATE_ADD(order_date, 7) AS expected_delivery_date
210 FROM shipping_orders;
```

Add parameter

← Results 3 of 3 →					
Table +					
	1 ² order_id	1 ² customer_id	📅 order_date	📅 expected_delivery_date	
1	6001	101	2026-05-01	2026-05-08	
2	6002	102	2026-05-03	2026-05-10	
3	6003	103	2026-05-05	2026-05-12	
4	6004	104	2026-05-07	2026-05-14	
5	6005	105	2026-05-09	2026-05-16	

Question 11: Year, Month, Day Separately

```
214
215 -- Q11: Year, Month, Day Separately
216 CREATE OR REPLACE TABLE bookings (
217     booking_id STRING,
218     customer_id INT,
219     booking_date DATE
220 );
221 INSERT INTO bookings VALUES
222     ('B001', 101, '2026-01-12'),
223     ('B002', 102, '2026-02-18'),
224     ('B003', 103, '2026-03-22'),
225     ('B004', 104, '2026-04-09'),
226     ('B005', 105, '2026-05-27');
227
228 SELECT
229     booking_id,
230     customer_id,
231     booking_date,
232     YEAR(booking_date) AS booking_year,
233     MONTH(booking_date) AS booking_month,
234     DAY(booking_date) AS booking_day
235 FROM bookings;
```

Add parameter

← Results 3 of 3 →						
Table +						
	1 ² booking_id	1 ² customer_id	📅 booking_date	1 ² booking_year	1 ² booking_month	1 ² booking_day
1	B001	101	2026-01-12	2026	1	12
2	B002	102	2026-02-18	2026	2	18
3	B003	103	2026-03-22	2026	3	22
4	B004	104	2026-04-09	2026	4	9
5	B005	105	2026-05-27	2026	5	27

Question 12: Filter Orders from 2026

```
236
237 -- Q12: Filter Orders from 2026
238 CREATE OR REPLACE TABLE yearly_orders (
239   order_id INT,
240   customer_id INT,
241   order_date DATE,
242   amount DECIMAL(10,2)
243 );
244 INSERT INTO yearly_orders VALUES
245   (7001, 101, '2024-12-15', 500),
246   (7002, 102, '2025-01-20', 1200),
247   (7003, 103, '2025-06-10', 800),
248   (7004, 104, '2026-02-05', 1500),
249   (7005, 105, '2026-05-25', 2000);
250
251 SELECT
252   order_id,
253   customer_id,
254   order_date,
255   YEAR(order_date) AS order_year,
256   amount
257 FROM yearly_orders
258 WHERE YEAR(order_date) = 2026;
```

Add parameter

Results 3 of 3					
Table					
	order_id	customer_id	order_date	order_year	amount
1	7004	104	2026-02-05	2026	1500.00
2	7005	105	2026-05-25	2026	2000.00

Question 13: Filter Orders from March

```
259
260 -- Q13: Filter Orders from March
261 CREATE OR REPLACE TABLE monthly_orders (
262   order_id INT,
263   customer_id INT,
264   order_date DATE,
265   amount DECIMAL(10,2)
266 );
267 INSERT INTO monthly_orders VALUES
268   (8001, 101, '2026-01-15', 500),
269   (8002, 102, '2026-02-20', 1200),
270   (8003, 103, '2026-03-10', 800),
271   (8004, 104, '2026-03-25', 1500),
272   (8005, 105, '2026-05-30', 2000);
273
274 SELECT
275   order_id,
276   customer_id,
277   order_date,
278   MONTH(order_date) AS order_month,
279   amount
280 FROM monthly_orders
281 WHERE MONTH(order_date) = 3;
282
```

Add parameter

Results 3 of 3					
Table					
	order_id	customer_id	order_date	order_month	amount
1	8003	103	2026-03-10	3	800.00
2	8004	104	2026-03-25	3	1500.00

Question 14: Last Day of the Month

```
282
283 -- Q14: Last Day of the Month
284 CREATE OR REPLACE TABLE subscriptions (
285   subscription_id STRING,
286   customer_id INT,
287   start_date DATE
288 );
289 INSERT INTO subscriptions VALUES
290   ('SUB001', 101, '2026-01-10'),
291   ('SUB002', 102, '2026-02-15'),
292   ('SUB003', 103, '2026-03-20'),
293   ('SUB004', 104, '2026-04-25'),
294   ('SUB005', 105, '2026-05-30');
295
296 SELECT
297   subscription_id,
298   customer_id,
299   start_date,
300   LAST_DAY(start_date) AS month_end_date
301 FROM subscriptions;
302
```

Add parameter

Results 3 of 3					
Table					
	order_id	customer_id	order_date	order_month	amount
1	8003	103	2026-03-10	3	800.00
2	8004	104	2026-03-25	3	1500.00

Question 15: First Day of the Month

```
303 -- Q15: First Day of the Month
304 CREATE OR REPLACE TABLE campaign_sends (
305   send_id STRING,
306   customer_id INT,
307   send_date DATE
308 );
309 INSERT INTO campaign_sends VALUES
310   ('C001', 101, '2026-01-12'),
311   ('C002', 102, '2026-02-18'),
312   ('C003', 103, '2026-03-22'),
313   ('C004', 104, '2026-04-09'),
314   ('C005', 105, '2026-05-27');
315
316 SELECT
317   send_id,
318   customer_id,
319   send_date,
320   DATE_TRUNC('MONTH', send_date) AS month_start_date
321 FROM campaign_sends;
322
```

Add parameter

Results 3 of 3				
Table				
send_id	customer_id	send_date	month_start_date	
C001	101	2026-01-12	2026-01-01T00:00:00.000+00:...	
C002	102	2026-02-18	2026-02-01T00:00:00.000+00:...	
C003	103	2026-03-22	2026-03-01T00:00:00.000+00:...	
C004	104	2026-04-09	2026-04-01T00:00:00.000+00:...	
C005	105	2026-05-27	2026-05-01T00:00:00.000+00:...	

Question 16: Format as Month and Year

```
327 -- Q16: Format as Month and Year
328 CREATE OR REPLACE TABLE invoice_dates (
329   invoice_id STRING,
330   customer_id INT,
331   invoice_date DATE
332 );
333 INSERT INTO invoice_dates VALUES
334   ('INV001', 101, '2026-01-05'),
335   ('INV002', 102, '2026-02-10'),
336   ('INV003', 103, '2026-03-15'),
337   ('INV004', 104, '2026-04-20'),
338   ('INV005', 105, '2026-05-25');
339
340 SELECT
341   invoice_id,
342   customer_id,
343   invoice_date,
344   DATE_FORMAT(invoice_date, 'MMM-yyyy') AS invoice_month_year
345 FROM invoice_dates;
346
347 -- Q17: Customer Age from Date of Birth
```

Add parameter

Results 3 of 3				
	invoice_id	customer_id	invoice_date	invoice_month_year
1	INV001	101	2026-01-05	January 2026
2	INV002	102	2026-02-10	February 2026
3	INV003	103	2026-03-15	March 2026
4	INV004	104	2026-04-20	April 2026
5	INV005	105	2026-05-25	May 2026

Question 17: Customer Age from Date of Birth

```
347 -- Q17: Customer Age from Date of Birth
348 CREATE OR REPLACE TABLE customer_birthdays (
349   customer_id INT,
350   customer_name STRING,
351   date_of_birth DATE
352 );
353 INSERT INTO customer_birthdays VALUES
354   (901, 'John', '1998-05-10'),
355   (902, 'Mary', '1990-08-20'),
356   (903, 'Peter', '2002-03-15'),
357   (904, 'Sarah', '1985-12-01'),
358   (905, 'Thabo', '2000-07-30');
359
360 SELECT
361   customer_id,
362   customer_name,
363   date_of_birth,
364   FLOOR(DATEDIFF(CURRENT_DATE(), date_of_birth) / 365.25) AS customer_age
365 FROM customer_birthdays;
366
367 -- Q18: Classify Weekend vs Weekday
```

Add parameter

Results 3 of 3				
	customer_id	customer_name	date_of_birth	customer_age
1	901	John	1998-05-10	28
2	902	Mary	1990-08-20	35
3	903	Peter	2002-03-15	24
4	904	Sarah	1985-12-01	40
5	905	Thabo	2000-07-30	26

Question 18: Classify Weekend vs Weekday

```
366
367 -- Q18: Classify Weekend vs Weekday
368 CREATE OR REPLACE TABLE weekend_orders (
369   order_id INT,
370   customer_id INT,
371   order_date DATE
372 );
373 INSERT INTO weekend_orders VALUES
374   (9001, 101, '2026-05-01'),
375   (9002, 102, '2026-05-02'),
376   (9003, 103, '2026-05-03'),
377   (9004, 104, '2026-05-04'),
378   (9005, 105, '2026-05-05');
379
380 SELECT
381   order_id,
382   customer_id,
383   order_date,
384   DAYNAME(order_date) AS day_name,
385   CASE
386     WHEN DAYNAME(order_date) IN ('Saturday', 'Sunday') THEN 'Weekend'
387     ELSE 'Weekday'
388   END AS day_type
389 FROM weekend_orders;
```

Add parameter

	order_id	customer_id	order_date	day_name	day_type
1	9001	101	2026-05-01	Fri	Weekday
2	9002	102	2026-05-02	Sat	Weekday
3	9003	103	2026-05-03	Sun	Weekday
4	9004	104	2026-05-04	Mon	Weekday
5	9005	105	2026-05-05	Tue	Weekday

Question 19: Transactions by Quarter

```
390
391 -- Q19: Transactions by Quarter
392 CREATE OR REPLACE TABLE quarterly_transactions (
393   transaction_id STRING,
394   customer_id INT,
395   transaction_date DATE,
396   amount DECIMAL(10,2)
397 );
398 INSERT INTO quarterly_transactions VALUES
399   ('Q001', 101, '2026-01-15', 500),
400   ('Q002', 102, '2026-03-20', 1200),
401   ('Q003', 103, '2026-04-10', 800),
402   ('Q004', 104, '2026-07-05', 1500),
403   ('Q005', 105, '2026-10-25', 2000);
404
405 SELECT
406   transaction_id,
407   customer_id,
408   transaction_date,
409   QUARTER(transaction_date) AS transaction_quarter,
410   amount
411 FROM quarterly_transactions;
```

Add parameter

	transaction_id	customer_id	transaction_date	transaction_quarter	amount
1	Q001	101	2026-01-15	1	500.00
2	Q002	102	2026-03-20	1	1200.00
3	Q003	103	2026-04-10	2	800.00
4	Q004	104	2026-07-05	3	1500.00
5	Q005	105	2026-10-25	4	2000.00

Question 20: Orders Older Than 30 Days

```
412
413 -- Q20: Orders Older Than 30 Days
414 CREATE OR REPLACE TABLE recent_orders (
415   order_id STRING,
416   customer_id INT,
417   order_date DATE,
418   amount DECIMAL(10,2)
419 );
420 INSERT INTO recent_orders VALUES
421   ('R001', 101, '2026-04-01', 500),
422   ('R002', 102, '2026-04-15', 1200),
423   ('R003', 103, '2026-05-01', 800),
424   ('R004', 104, '2026-05-10', 1500),
425   ('R005', 105, '2026-05-25', 2000);
426
427 SELECT
428   order_id,
429   customer_id,
430   order_date,
431   DATEDIFF(CURRENT_DATE(), order_date) AS days_since_order,
432   amount
433 FROM recent_orders
434 WHERE DATEDIFF(CURRENT_DATE(), order_date) > 30;
```

Add parameter

← Results 3 of 3 →

	order_id	customer_id	order_date	days_since_order	amount
1	R001	101	2026-04-01	122	500.00
2	R002	102	2026-04-15	108	1200.00
3	R003	103	2026-05-01	92	800.00
4	R004	104	2026-05-10	83	1500.00
5	R005	105	2026-05-25	68	2000.00

Bonus Question

```
441 CREATE OR REPLACE TABLE customer_recency (
442   customer_id INT,
443   customer_name STRING,
444   last_purchase_date DATE,
445   total_spend DECIMAL(10,2)
446 );
447 INSERT INTO customer_recency VALUES
448   (1001, 'John', '2026-05-25', 5000),
449   (1002, 'Mary', '2026-05-10', 2500),
450   (1003, 'Peter', '2026-04-01', 700),
451   (1004, 'Sarah', '2026-02-15', 15000),
452   (1005, 'Thabo', '2025-12-20', 300);
453
454 SELECT
455   customer_id,
456   customer_name,
457   last_purchase_date,
458   DATEDIFF(CURRENT_DATE(), last_purchase_date) AS days_since_last_purchase,
459   CASE
460     WHEN DATEDIFF(CURRENT_DATE(), last_purchase_date) <= 30 THEN 'Active Customer'
461     WHEN DATEDIFF(CURRENT_DATE(), last_purchase_date) BETWEEN 31 AND 90 THEN 'At-Risk Customer'
462     ELSE 'Inactive Customer'
463   END AS customer_status
464 FROM customer_recency;
```

Add parameter

← Results 3 of 3 →

	customer_id	customer_name	last_purchase_date	days_since_last_purchase	customer_status
1	1001	John	2026-05-25	68	At Risk Customer
2	1002	Mary	2026-05-10	83	At Risk Customer
3	1003	Peter	2026-04-01	122	Inactive Customer
4	1004	Sarah	2026-02-15	167	Inactive Customer
5	1005	Thabo	2025-12-20	224	Inactive Customer